

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

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COURSE CODE: ITA0612

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.

(5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.

(5 Marks)

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5

Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather,

Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example. (5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

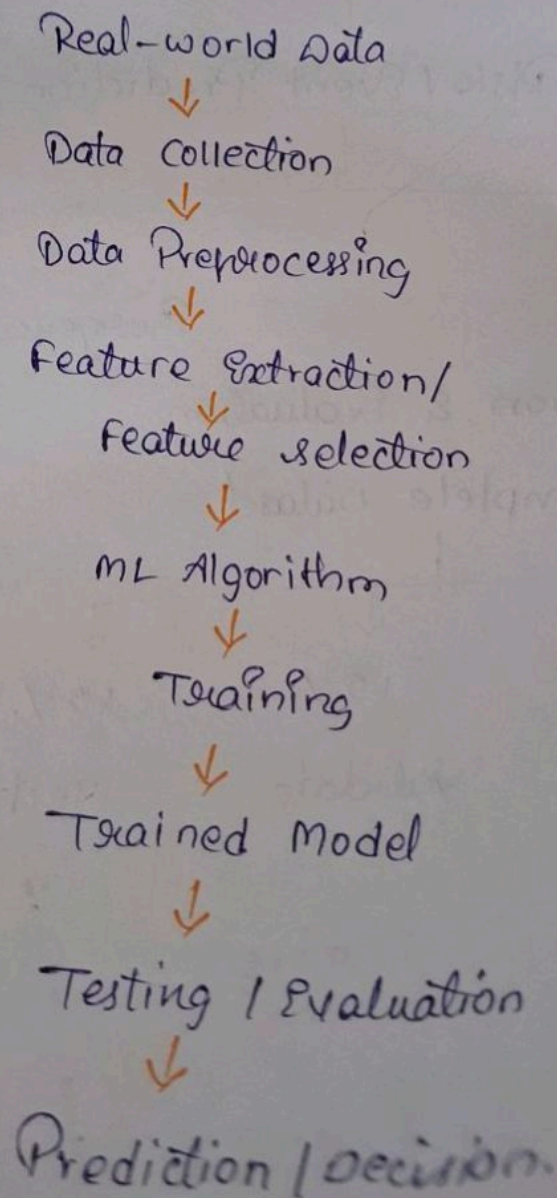
Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Machine Learning① Machine Learning system

A machine learning (ML) system learns patterns from data and uses the learned knowledge to make Predictions or decisions without being Explicitly programmed for Every case.



2) Early warning and Emergency Response

Real-time sensors



Data processing



ML model



Risk level



warning Alert



People / Authorities

3) Find-S Algorithm - Play Tennis

$H_0 = \langle \phi, \phi, \phi, \phi \rangle$

Example 3 = $\langle \text{overcast}, \text{Hot}, \text{High}, \text{weak} \rangle$

$H_1 = \langle \text{overcast}, \text{Hot}, \text{High}, \text{weak} \rangle$

Example 4 = $\langle \text{Rainy}, \text{mild}, \text{High}, \text{weak} \rangle$

Outlook differs $\rightarrow ?$

Temperature differs $\rightarrow ?$

Humidity same $\rightarrow \text{High}$

wind same $\rightarrow \text{weak}$

$\therefore$

$H_2 = \langle ?, ?, \text{High}, \text{weak} \rangle$

$H_2 = \langle ?, ?, \text{High}, \text{weak} \rangle$

Example 5 = $\langle \text{Rainy}, \text{cool}, \text{Normal}, \text{weak} \rangle$

② Machine Learning system for Disaster management

Historical Data

Real-time Data.

Data preprocessing

Feature selection

ML Prediction model

Risk / Event Prediction

Early warning

Emergency Response.

a) Training, Validation & Evaluation

Complete dataset

40%

15%

15%

Train

validate

Test

Learning

Tuning

Final

Evaluation

$$H_3 = \langle ?, ?, ?, \text{weak} \rangle$$

$$\text{Example } t = \langle \text{overcast}, \text{mild}, \text{Normal}, \text{strong} \rangle$$

$$H_4 = \langle ?, ?, ?, ? \rangle$$

$$H = \langle ?, ?, ?, ? \rangle$$

4. Candidate Elimination Algorithm - Enjoy Sport

$\langle \text{weather}, \text{Temperature}, \text{Humidity}, \text{wind} \rangle$

we maintain two boundaries:

- S = specific boundary
- G = General boundary

Initial

$$S_0 = \langle \phi, \phi, \phi, \phi \rangle$$

$$G_0 = \langle ?, ?, ?, ? \rangle$$

Sunny, warm, Normal, weak $\rightarrow$ yes

$$S_1 = \langle \text{Sunny}, \text{warm}, \text{Normal}, \text{weak} \rangle$$

$$G_1 = \{ \langle ?, ?, ?, ? \rangle \}$$

$$S = \langle \text{Sunny}, \text{warm}, \text{Normal}, \text{weak} \rangle$$

$$G_2 = \langle \text{Sunny}, \text{warm}, \text{High}, \text{strong} \rangle$$

weather same $\rightarrow$ Sunny

Temperature same $\rightarrow$ warm

Humidity differs $\rightarrow ?$

wind differs $\rightarrow ?$

(2)

① Explore:

$$S_2 = \langle \text{Sunny}, \text{warm}, ?, ? \rangle$$

 G_1 Remains:

$$G_{12} = \{ \langle ?, ?, ?, ? \rangle \}$$

Rainy, Cold, High, weak $\rightarrow$ No

$$G_1 = \langle ?, ?, ?, ? \rangle$$

 $\langle \text{Sunny}, ?, ?, ? \rangle$ $\langle ?, \text{warm}, ?, ? \rangle$ $\langle ?, ?, \text{Normal}, ? \rangle$ $\langle ?, ?, ?, \text{strong} \rangle$

$$G_3 = \{ \langle \text{Sunny}, ?, ?, ? \rangle, \langle ?, \text{warm}, ?, ? \rangle \}$$

$$S_3 = \langle \text{Sunny}, \text{warm}, ?, ? \rangle$$

$$P_4 = \langle \text{Sunny}, ?, ?, ? \rangle$$

$$G_5 = \{ \langle \text{Sunny}, ?, ?, ? \rangle \}$$

$$\text{version space} = \{ \langle \text{Sunny}, ?, ?, ? \rangle \}$$

5. Find-S algorithm - Customer Buys Product $\langle \text{Age}, \text{Income}, \text{student}, \text{Credit Ratings} \rangle$

$$H_0 = \langle \phi, \phi, \phi, \phi \rangle$$

Young, High, No, Fair $\rightarrow$ No

$$\therefore H_0 = \langle \phi, \phi, \phi, \phi \rangle$$

middle, High, No, fair $\rightarrow$ Yes

$$H_1 = \langle \text{middle, High, No, fair} \rangle$$

$$H_1 = \langle \text{middle, High, No, fair} \rangle$$

$$H_4 = \langle \text{Senior, medium, No, fair} \rangle$$

$$H_2 = \langle ?, ?, \text{No, fair} \rangle$$

$$H = \langle ?, ?, \text{No, fair} \rangle$$